Advanced Database
Database 2

YPU 2024-2025

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Course Outline
Advanced Concepts in DBMS

1. *Database Management Systems - Review*
2. *ER model - Review*
3. *Relational Algebra - Review*
4. *Stored Procedures*
5. *Functions*
6. *Triggers*
7. *Normalization*
8. *Indexing*
9. *Query Processing*
10. *Query Optimization*
11. *Transactions*
12. *Handling failures*
13. *Concurrency Control*

1. *Database Management Systems Review*

Chapter 1-a

What Is a DBMS?

- **A very large, integrated collection of data.**
- **Models real-world *enterprise.***
  - **Entities (e.g., students, courses)**
  - **Relationships (e.g., Ahmad is taking DB2)**
- **A *Database Management System (DBMS)* is a software package designed to store and manage databases.**

Why Use a DBMS?

- **Data independence and efficient access.**
- **Reduced application development time.**
- **Data integrity and security.**
- **Uniform data administration.**
- **Concurrent access, recovery from crashes**.

Why Study Databases?

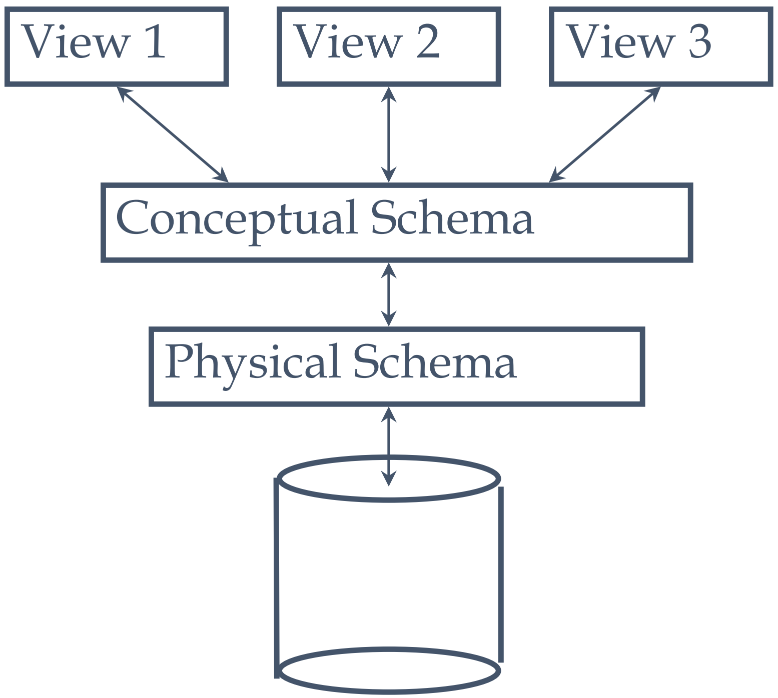
- Shift from *computation* to *information*
  - at the “low end”: scramble to webspace (a mess!)
  - at the “high end”: scientific applications
- Datasets increasing in diversity and volume.
  - Digital libraries, interactive video, Human Genome project
  - Need for DBMS exploding
- DBMS encompasses most of CS
  - OS, languages, theory, AI, multimedia, logic

Data Models

- A *data model* is a collection of concepts for describing data.
- A *schema* is a description of a particular collection of data, using a given data model.
- The *relational model of data* is the most widely used model today.
  - Main concept: *relation*, basically a table with rows and columns.
  - Every relation has a *schema*, which describes the columns, or fields.

Levels of Abstraction

- Many *views*, single *conceptual (logical) schema* and *physical schema*.
  - Views describe how users see the data.
  - Conceptual schema defines logical structure
  - Physical schema describes the files and indexes used.



Schemas are defined using DDL; data is modified/queried using DML.

Example: University Database

- Conceptual schema:
  - *Students(sid: string, name: string, login: string,*

    age: integer, gpa:real)
  - *Courses(cid: string, cname:string, credits:integer)*
  - *Enrolled(sid:string, cid:string, grade:string)*
- Physical schema:
  - Relations stored as unordered files.
  - Index on first column of Students.
- External Schema (View):
  - *Course_info(cid:string,enrollment:integer)*

Data Independence

- Applications insulated from how data is structured and stored.
- *Logical data independence*: Protection from changes in *logical* structure of data.
- *Physical data independence*: Protection from changes in *physical* structure of data.

One of the most important benefits of using a DBMS!

Concurrency Control

- Concurrent execution of user programs is essential for good DBMS performance.
  - Because disk accesses are frequent, and relatively slow, it is important to keep the cpu humming by working on several user programs concurrently.
- Interleaving actions of different user programs can lead to inconsistency: e.g., check is cleared while account balance is being computed.
- DBMS ensures such problems don’t arise: users can pretend they are using a single-user system.

Structure of a DBMS

- A typical DBMS has a layered architecture.
- The figure does not show the concurrency control and recovery components.
- This is one of several possible architectures; each system has its own variations.